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Project practice 2025/2026

How to write an amazing article

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Abstract
What is the problem? What is the topic? The aim of this paper? Lorem ipsum dolor sit amet, consectetur adipiscing elit. Fusce ullamcorper suscipit euismod. Mauris sed lectus non massa molestie congue. In hac habitasse platea dictumst. How is the problem solved, the aim achieved (methodology)? Lorem ipsum dolor sit amet, consectetur adipiscing elit. Fusce ullamcorper suscipit euismod. Mauris sed lectus non massa molestie congue. In hac habitasse platea dictumst. Curabitur massa neque, commodo posuere fringilla ut, cursus at dui. Nulla quis purus a justo pellentesque. What are the specific results? How well is the problem solved? Lorem ipsum dolor sit amet, consectetur adipiscing elit. Fusce ullamcorper suscipit euismod. Mauris sed lectus non massa molestie congue. In hac habitasse platea dictumst. So what? How useful is this to Science and to the reader? Lorem ipsum dolor sit amet, consectetur adipiscing elit. Fusce ullamcorper suscipit euismod.

Keywords: Keyword1 — Keyword2 — Keyword3

Supplementary Material: Demonstation video — Downloadable code

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1. Introduction

With the rise of Large Language Models (LLMs) and the increase in their capabilities, companies are motivated to use LLMs for automation tasks, including testing and its tooling. However, LLMs are not plug-and-play; specialized tooling is often required to produce reliable outputs. This work defines a formal base into which LLMs rewrite requirements and test cases written in natural language, so that the result is parsable by coverage checking systems.

There is an increasing demand to instrument systems in natural language, including the specification of requirements and their test cases for testing systems and system components. Even when requirements and test cases are written in English, they must still be rewritten into a mathematical formal base so that existing test-coverage techniques can be applied. A formal base into which LLMs can convert human-written text is therefore needed. This formal base must be able to express all motivational requirement examples. Its semantics must be easily interpretable by LLMs and agentic systems, so that agents can rewrite requirements and test case descriptions from human language into the formal base with high accuracy and without changing the semantics; the formalism itself must be unambiguous. It must also be convertible to existing formal bases used by formal verification tools. A good formalism should meet all of the mentioned criteria.

2. Existing methods of requirement formalisation – State of the Art

3. Formalism requirements & example use cases

4. Formalism design decisions

5. Known limitations and possible future extensions

6. Compatibility with well-known formal specification methods

7. Requirement and test case examples

8. Future work

9. Conclusions

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References